

Danial Kamali

✉ kamalida@msu.edu | 🏠 iamdanialkamali.github.io | 📷 iamdanialkamali

Education

Michigan State University

PHD IN COMPUTER SCIENCE (GPA: 4.0, RESEARCH: VISION & LANGUAGE)

East Lansing, Michigan, USA

Aug 2022 - Present

Iran University of Science and Technology

B.Sc IN COMPUTER ENGINEERING (RESEARCH: FACIAL EXPRESSION RECOGNITION, NLP LEXICAL ANALYSIS)

Tehran, Iran

Sept 2016 - March 2021

Experiences

Michigan State University

GRADUATE RESEARCH ASSISTANT

East Lansing, USA

Aug 2022 - Present

- Implemented various deep learning tasks in DomiKnowS (A Library for Integration of Symbolic Domain Knowledge in Deep Learning) for the paper.
- Researching Compositional Generalization on multi-modal systems improves generalization in unseen structures.
- Lead a team of researchers on data gathering, labeling, and baseline preparation on the misinformation detection task.

Zibal

BACKEND TEAM LEADER

Tehran, Iran

Aug 2021 - Aug 2022

- Managed a team of 4 people with the Agile manifesto and improved the team capacity utilization.
- Reviewed backend team merge requests on Gitlab and reduced the defection rate.
- Mentored two interns through 8 weeks of training, one of them joined our team as a junior developer.

FULL-STACK DEVELOPER

Aug 2021 - Aug 2022

- Stack:** ReactJS, Ant Design, LESS.

BACKEND DEVELOPER

Oct 2019 - Aug 2021

- Stack:** Django, DRF, TensorFlow2, MongoDB, Postgres, Redis, Celery, Docker, Kubernetes
- Developed data aggregation pipelines for BI system from NOSQL to SQL-based DB for the product team.
- Implemented Django-MongoDB generic view leveraging reusable patterns to increased developer productivity.
- Designed CI/CD using Gitlab and kubernetes for front-end and back-end and automated the developemnt and deploy operations.

IUST Data Mining Lab

NATURAL LANGUAGE PROCESSING RESEARCH ASSISTANT

Tehran, Iran

Jun 2019 - Dec 2021

- Evaluated Farsi Sentences Tokenization and analyzed current Farsi tokenizers.
- Mentored undergrad thesis projects and provided feedback on different stages of their project.

IUST Multi-Agent Systems Lab

MACHINE LEARNING RESEARCH ASSISTANT

Tehran, Iran

Jun 2019 - Sept 2019

- Researched on Image Caption Generation and customized image captioning in Farsi.
- Gathered the Farsi image captioning dataset using the coco dataset.
- Trained the model using the attention mechanism on the mentioned dataset, using TensorFlow.

Publications

- Kamali, Danial**, Barezi, E. J., & Kordjamshidi, P. "NeSyCoCo: A Neuro-Symbolic Concept Composer for Compositional Generalization." *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025.
- Kamali, Danial**, Romain, J., Liu, H., Peng, M., & Kordjamshidi, P. "Misinformation Detection Using Persuasive Writing Strategies." *Joint International Conference on Computational Linguistics, Language Resources, and Evaluation (COLING)*, 2024.
- Kamali, Danial**, & Kordjamshidi, P. "Syntax-Guided Transformers: Elevating Compositional Generalization and Grounding in Multimodal Environments." *GenBench: Workshop on Generalization Benchmarking in NLP, Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Kamali, Danial**, Janfada, B., Shensa, E., & Minaie Bidgoli, B. "Evaluation of Persian Tokenizers." *Preprint*, under review.

Skills

Programming	Python, Java Script, C/C++, JAVA, C#		
AI Frameworks	Sklearn, Tensorflow, PyTorch , Keras, HuggingFace, Numpy, OpenCV, NLTK		
Tools	Kubernetes, Docker , MYSQL, MongoDB , Redis, Celery , Kafka		
Frameworks and Libraries	Django , DRF, React, Express, Flask, Unity		
Language	Persian:	<i>MotherTongue</i>	English: TOEFL IBT test score: 105 (R:29 L:29 S:23 W:24)

Projects

Neural Machine Translation

- Initiated the development of English ol'Saltalneh, an NMT model for Farsi-to-Finglish transliteration, enhancing translation for English forms.
- Curated a dataset of 100,000+ Farsi-English samples, leading data-cleaning efforts to standardize input and improve model training outcomes.
- Devised and applied data augmentation techniques to bolster the dataset, resulting in a 20% BLEU score enhancement for the model.

Facial Expression Recognition

- Developed a convolutional neural network (CNN) that achieved state-of-the-art accuracy on the RAF-DB dataset and competitive results on FER2013.
- Implemented data augmentation and transfer learning techniques to improve model generalizability.
- Implemented statistically aware layers to the network, increasing the interpretability of facial feature representations.

Memotion Analysis: MultiModal Image-Text classification

- Engaged in Memotion Analysis competition on CodaLab, developing a multimodal classification system using NLP and Computer Vision.

Real-time Augmented Reality Sudoku Solver

- Engineered a real-time AR application that captures Sudoku puzzles via video stream, solves them, and overlays the solutions.
- Implemented a custom digit recognition algorithm using neural networks that achieved 95% accuracy on handwritten digits.
- Developed the AR overlay system using OpenCV and Unity, resulting in an interactive and user-friendly application experience.

Awards

2022	Recipient Dr. Carl Page Sr Fellowship	MSU
2020	Ranked 1st Annual evaluation of CE department students	IUST
2015	Ranked among the top 0.2% of the candidates National Entrance Exam for Graduate Schools	Iran
2013, 2014	Qualified for the second phase National Mathematics Olympiad	Iran
2013	Qualified for Iran national team International Mathematics Competition	Iran
2012	Exceptional talent scholarship Salam high school	Iran

Services & Activities

2024	Student Volunteer EMNLP’24 GenBench Workshop	MSU
2023-2024	Reviewer ACL’23, IJACI’24, EMNLP’23	MSU
2023-2024	President Iranian Students Association	MSU
2022	Treasurer Iranian Students Association	MSU
2022	Development Team GAMEIN Event	Sharif University
2020-2021	Undergraduate Project Mentor Data Mining Lab	IUST

Teaching Experience

TA	Computer Organization and Architecture	May 2024 - Aug 2024
TA	Introduction to Programming	Jan 2023 - May 2023
TA	Computer Vision	Sep 2020 - Feb 2021
TA	Computational Intelligence	Feb 2020 - Aug 2020
Instructor	Natural Language Processing	Feb 2020 - Aug 2020
Mentor	Software Engineering	Sep 2019 - Jan 2020
TA	Computer Architecture	Feb 2019 - Jun 2019
Mentor	Software Engineering	Jan 2019 - May 2019
TA	Database Design	Jan 2019 - May 2019
TA	Digital Logic	Sep 2018 - Jan 2019

Selected Academic Courses

MSU	IUST
• Deep Learning: 4.0 (top 3) • Computational Foundation of AI: 4.0 (top 1) • Natural Language Processing: 4.0 • Advanced Algorithm Design: 4.0	• Computer Vision: 4.0 • Deep Learning: 4.0 • Computational Intelligence: 4.0 • Natural Language Processing: 4.0